Round 5 desk work: CHW body, a description-length bound for low-degree tests, literature (2026-11-05)

This report records reading and derivation work, plus one exact toy check (h1pp_check.py, h1pp_scaling.py, [pcc:unverified] about 30 s on one core). It contains no timing measurements and makes no cost claims. Every statement is tagged as read, proved, cited or heuristic in the two linked notes.

1. KN-LIT-7606 (structured generic-group model): body read

The full note is chw-reading.md. The PDF is the author copy, sha256 82b40c27…6af66f.

- **There is no elliptic-curve theorem in the body.** The abstract mentions "elliptic-curve points", but §5.3 lists elliptic-curve special-case algorithms as not yet modelled. The paper's applications are to smooth and small integers and to smooth polynomials only.
- **What the bound covers.** Theorem 3.2 gives DLAdv ≤ δn(3T+2)/q + (3T+1)²/q + 1/n, where δ is the fraction of labels that occur in a ⋆ relation. The proof fixes the labelling on that δ-fraction, which the adversary may know in full, and randomises everything else.
- **What follows for coordinates.** By a reading of the proof that the authors do not state ([pcc:unverified]):
  - Coordinate access confined to a fixed set, such as a coordinate-defined factor base of B ≤ √q points, gives no advantage over the generic bound.
  - Index calculus reads coordinates of instance-derived points. There δ = 1 and the bound is vacuous. The authors say the same for integers (§7).
  - So there is no query-model closure for full coordinate access. LEAD-EXO-coordinate-revealing-oracles step (2) is settled.
- **Correction to the corpus.** KN-LIT-7606 earlier said that CHW "already covers elliptic-curve point structure" and is "direct prior art" closing the KN-FIND-002 oracles. Both statements are withdrawn in the record. This is input to review round BATCH-9f090a (joint J2). It changes no finding's status here.

2. LEAD-EXO-low-degree-query-lower-bound, step (c): charging the description length

The full note is h1pp-note.md. It was written by an Opus agent and checked here. Both JSON outputs were reproduced byte for byte by a rerun.

- **H1″ (proved).** Pr[success] ≤ (C(T_g+2,2) + 1 + Σ_j(μ_j + t_j))/N, where μ_j is the smaller side of test j's [pcc:unverified] split of the candidates. The simulator answers each test with its majority answer. H1′ (the Delta bound) is the special case μ ≤ Delta.
- **Floor (proved).** An L-monomial test can vanish on any L − 1 chosen candidates, so no description-charged bound can do better than Σ L_j.
- **Counting (proved).** Cutting out a prescribed generic balanced set needs Ω(N / log N) monomials, or straight-line-program size Ω(N / log pN). This is the rigorous form of HD2c. It gives no worst-case bound, because the solver chooses its own sets.
- **Conjecture H1″-strong (heuristic).** On a prime-order curve, every L-monomial test has μ ≤ L·N^o(1). That would force T_g ≳ √N or Σ L_j ≳ N^(1−o(1)). It is open even for L = 2: Weil and Kohel–Shparlinski scale with Delta and are vacuous for large multipliers.
- **Prime order matters (proved).** Over F_p^*, the binomial $x_1 - x_2$ with $b_2 = b_1 + (p-1)/2$ splits the candidates in half.

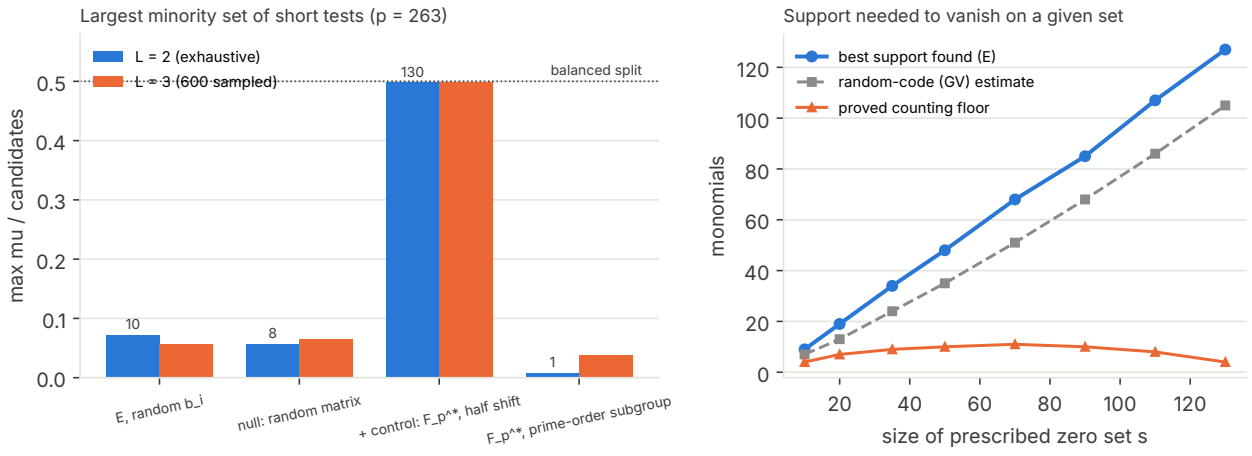

Figure 1: Largest minority set of short tests

arm (p = 263, N = 281, t = 9, D = 3)	candidates	rank	max μ, L = 2 (exhaustive)	max μ, L = 3 (600 sampled)
E, random multipliers	140	140	10	8
null: uniform random matrix	140	140	8	9
positive control: F_263^*, $b_2 = b_1 + 131$	261	157	130	130
F_263^*, prime-order subgroup (q = 131)	130	104	1	5

At toy size the curve behaves like the null: no short balanced test exists. This is consistent with H1″-strong and is not evidence for its asymptotics. The row stays open_next_step. Its next step is a desk bound on the largest fiber of ∏ x([b_i]R)^(e_i) for t = 2.

3. Literature

- **Kiltz–Winterhof** (Bull. Austral. Math. Soc. 69 (2004) 305–315; open Cambridge PDF): read in full.
  - It is about the Diffie–Hellman map, not the discrete log.
  - Its elliptic theorems (5–6) need agreement on more than half of the square, or on a near-full product set.
  - It does not decide A(d) for LEAD-EXO-invertible-hint-subset.
- **Lange–Winterhof** (COCOON 2002; AAECC 2003): not reachable by an open route (Springer paywall, author pages without PDFs).
- **Zhang–Liu** (ePrint 2018/795): the PDF returned 403 to automated clients. Only the first author's CAB Forum 2019 slides were read. They say the method is "not even of square-root time". The row records this as consistent with enumeration in disguise, not as a proof of it.
- No access control was bypassed, and TLS verification stayed on throughout.

Reproduce

```
python3 h1pp_check.py; python3 h1pp_scaling.py     # writes h1pp_check.json, h1pp_scaling.json (~30 s)
python3 figures.py; ./build_pdf.sh
```

h1pp_check.py's docstring says p = 251, but the script runs p = 263, as the JSON shows. The scripts are kept unchanged so that their hashes match the note.